Razonamiento Automático

GRADO EN INGENIERÍA EN INFORMÁTICA

Sesión 05.2

Introducción a ECS

2026-27

Challenge 4: ¿Qué falta?

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- ¿**10 platos** + **3 disparos**?
  - `clay1_row`, `clay2_row`… `clay10_row`
  - 10x `system_move_clay`

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  - `clay1_row`, `clay2_row`… `clay10_row`
  - 10x `system_move_clay`
- ***No escala***
  - Necesario ***organizar los datos***

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Entity

Simulable: enemigo, disparo, plato...

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Component

Datos: posición, tile, velocidad, vida...

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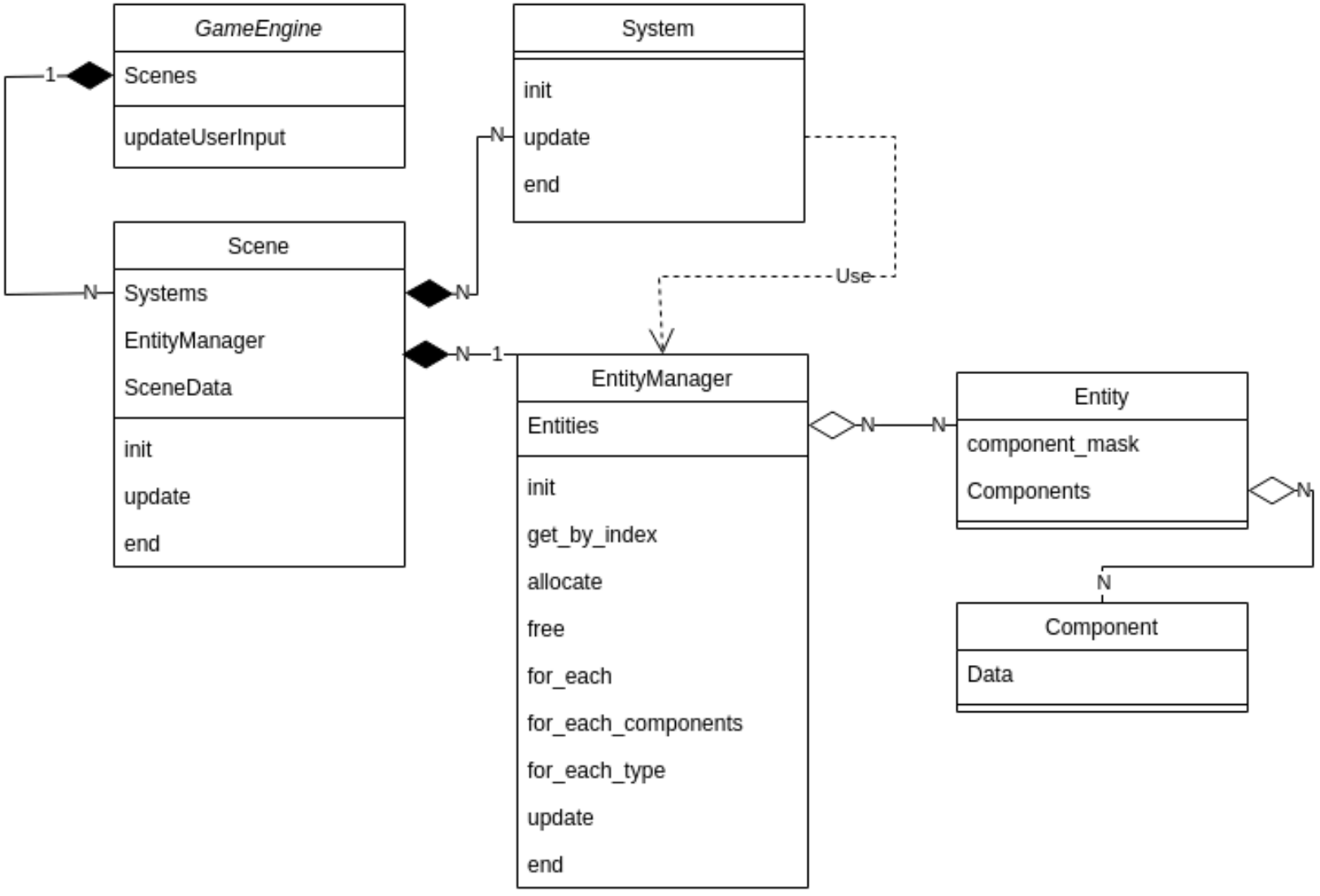
Component

Datos: posición, tile, velocidad, vida...

System

Proceso (código): mover, dibujar, detectar...

Diagrama ER básico



Entidades en memoria

Entities – continuos components

Address	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
C000	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
C010	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
C020	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
C030	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
C040	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
C050	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
C060	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
C070	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

One array
-
Entity is a single data Structure

Una entidad = un struct

```
;; struct Entity { active, y, x, tile }
RSRESET
DEF E_ACTIVE      RB 1   ;; 0 = free, 1 = in use
DEF E_Y           RB 1
DEF E_X           RB 1
DEF E_TILE        RB 1
DEF SIZEOF_ENTITY RB 0   ;; = 4
```

- RSRESET/RB: ***desplazamientos*** (0, 1, 2, 3)
- Campo = ***Base + desplazamiento***

```
ld hl, entity  ;; HL = base
ld bc, E_TILE  ;; BC = desplazamiento
add hl, bc     ;; HL = base + desplazamiento
```

Array de entidades

```
DEF MAX_ENTITIES = 8

SECTION "Entities", WRAM0, ALIGN[8]
entity_array: DS MAX_ENTITIES * SIZEOF_ENTITY   ;; 32 bytes

;; ACCEDER A LA ENTIDAD 'B'
   ld c, SIZEOF_ENTITY
   call multiply_b_by_c  ;; BC = B * C (multiplica)
   ld  hl, entity_array  ;; HL = Base
   add hl, bc            ;; HL += Desplazamiento
```

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- `ALIGN[8]`: Empieza en `$xx00`

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   call multiply_b_by_c  ;; BC = B * C (multiplica)
   ld  hl, entity_array  ;; HL = Base
   add hl, bc            ;; HL += Desplazamiento
```

- `ALIGN[8]`: Empieza en `$xx00`
- 256 bytes sin cambiar la parte alta

ECS: Entity Manager

Address	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	
C000	01	20	18	FF	00	BA	05	70	00	18	14	01	11	D5	03	70	Space for 10 Entities (8 bytes Each)
C010	00	10	0A	02	22	D5	02	30	07	17	16	FE	22	CD	06	00	
C020	17	22	10	FE	03	BC	05	70	00	00	00	00	00	00	00	00	
C030	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
C040	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
C050	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
C060	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	

```
SECTION "Entity Manager CODE", ROM0
man_entity_init:      ;; All entities free
man_entity_alloc:     ;; Finds a free one and marks it used
man_entity_free:      ;; Marks one as free
man_entity_for_each:  ;; Calls a routine for every active one
```

Un sistema

```
;; Moves ONE entity one row down
;; 📥 DE = entity
sys_physics_one:
   ld  h, d
   ld  l, e           ;; HL = entity
   ld  de, E_Y
   add hl, de         ;; HL = entity.E_Y
   inc [hl]           ;; y += 1
   ret

;; Moves ALL active entities
sys_physics_update::
   ld  hl, sys_physics_one
   jp  man_entity_for_each   ;; Tail-call (call+ret)
```

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- El sistema solo sabe procesar ***una*** entidad

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   add hl, de         ;; HL = entity.E_Y
   inc [hl]           ;; y += 1
   ret

;; Moves ALL active entities
sys_physics_update::
   ld  hl, sys_physics_one
   jp  man_entity_for_each   ;; Tail-call (call+ret)
```

- El sistema solo sabe procesar ***una*** entidad
- El gestor sabe ***recorrer*** todas

ECS: Scene

```
;; Initializes the game scene (Systems, data, etc)
scene_game_init:
    call man_entity_init
    call sys_render_init
    ;; ....
    ret
;; Updates the scene
scene_game_update:
    call sys_input_update
    call sys_physics_update
    call sys_collisions_update
    call sys_render_update
    ret
```

Código de ejemplo

⚠️ ***not production-ready!***

ECS: Jerarquía mental

Arquitectura ECS-GB

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Arquitectura ECS-GB

- Game-Engine
  - Scenes
    - Systems
    - Entity-Manager
      - Components<->Entities

Optimizando

Entities – sparse components

Address	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	
C000	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
C010	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	***Sprite*** Component
C020	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
C030	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
.....																	
C100	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
C110	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	***Physics*** Component
C120	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
.....																	
C200	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
C210	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	***AI*** Component *Part 1*
C220	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
.....																	
C300	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
C310	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	**AI** Component *Part 2*
C320	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	

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